Hasan Asif — Resume

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AI & Automation Engineer · Agentic Systems, Voice AI, RAG

Mirpur, AJK, Pakistan · +92 300 0993563 · hassanasif2003@gmail.com · linkedin.com/in/hasan-asif-293b412a2

Profile

AI and automation engineer with 2.5 years of hands-on experience building production LLM systems — autonomous agents, voice agents, agentic RAG pipelines, and LLM output verification. Currently Senior AI Engineer at SelfTech Pvt Ltd, where I joined as an intern and now lead AI delivery on commercial products including Smart Learning Hub, a UK-market AI tutoring platform. Software engineering degree from MUST, with end-to-end ownership across retrieval design, agent orchestration in n8n, and deployment on AWS, Azure, and Vercel. Alongside AI work, I build real-time interactive systems — including BlackZone, a 3D first-person shooter in Unity. Ship fast using AI-native development tooling (Claude Code, Codex).

Core Skills

- **Agentic AI:** Autonomous agent design, multi-agent orchestration, tool and function calling, agent memory and state management, human-in-the-loop checkpoints, guardrails and fallback handling
- **RAG & Retrieval:** Agentic RAG pipelines, ChromaDB, chunking and embedding strategy, semantic search, query rewriting, source-grounded answering, hallucination reduction
- **Voice AI:** Voice agent pipelines, real-time STT/TTS, turn-taking and latency optimisation, call handling, transcript capture and post-call automation
- **LLM Verification:** LLM-as-judge evaluation, output validation and schema enforcement, factual-consistency checks, automated quality gates before user-facing delivery
- **Automation:** n8n Cloud (agent workflows, webhooks, scheduled pipelines, API integrations), event-driven architecture, third-party system integration
- **Models & APIs:** Anthropic Claude, prompt engineering, context engineering, structured outputs, token and cost optimisation, model routing
- **Game Development:** Unity (3D, first-person mechanics, scene and asset design, gameplay scripting in C#), real-time interactive systems

- **Engineering:** Python, JavaScript/TypeScript, C#, REST APIs, web application development, Flutter (working knowledge)
- **Cloud & DevOps:** AWS, Microsoft Azure, Vercel, Docker, CI/CD, production deployment and monitoring
- **AI-Assisted Dev:** Claude Code, OpenAI Codex — spec-driven development, agentic refactoring, automated testing

Professional Experience

Senior AI Engineer — SelfTech Pvt Ltd

[Month] 2024 – Present

Promoted from Software Engineering Intern to Senior AI Engineer; lead AI engineer on commercial client products.

- Lead AI engineering on **Smart Learning Hub** (smartlearninghub.co.uk), a commercial UK-market AI tutoring platform covering GCSE and iGCSE Maths, Sciences, English and Literature — built *Professor Ellis*, the conversational AI tutor that delivers personalised explanations, adaptive learning paths and 24/7 student support.
- Designed and shipped **agentic RAG systems** over curriculum and document corpora using ChromaDB, with source-grounded retrieval and query rewriting to keep tutoring answers accurate and citable.
- Built **LLM verification systems** that validate model output before it reaches users — schema enforcement, factual-consistency checks and automated quality gates — reducing incorrect responses in production.
- Developed **voice agents** with real-time speech pipelines, including an AI voice assistant for customer support (see Selected Projects), optimising turn-taking and latency for natural conversation.
- Orchestrated multi-step AI agents and business automations in **n8n Cloud**, integrating LLM calls with external APIs and internal systems to replace manual workflows.
- Owned deployment and scale: containerised services with **Docker** and deployed across **AWS, Azure and Vercel**, with monitoring and cost controls on LLM inference spend.
- Built supporting web applications and interactive learning games as part of full product delivery, working across frontend, backend and AI layers.
- Accelerated delivery using **Claude Code and Codex** for spec-driven development, refactoring and test generation.

Selected Projects

Smart Learning Hub — AI Tutoring Platform

Commercial product, SelfTech Pvt Ltd · smartlearninghub.co.uk

AI-powered personalised learning platform serving UK GCSE and iGCSE students, with an AI tutor, adaptive learning paths and real-time progress analytics. Built the AI layer end to end: conversational tutoring agent, retrieval over curriculum content with ChromaDB, and verification of generated explanations. Deployed to production on AWS / Vercel.

Stack: Anthropic Claude, ChromaDB, Python, Node.js, n8n, AWS, Vercel.

AI Voice Assistant for Customer Support

Client / Internal · 2025

Production voice agent that handles inbound customer-support calls end to end — greeting, intent detection, knowledge-base lookup, resolution, and warm handoff to a human when needed. Built the real-time STT → LLM → TTS pipeline with turn-taking and barge-in support to keep conversation natural, grounded answers through agentic RAG over the client's support documentation, and post-call automation in n8n for ticket creation, transcript storage and CRM updates. Cut average handle time and freed human agents to focus on complex cases.

Stack: Anthropic Claude, real-time STT/TTS, ChromaDB, n8n Cloud, Python, webhook integrations.

BlackZone — 3D First-Person Shooter

Personal project · Unity

Full 3D first-person shooter built from scratch in Unity. Designed and implemented core FPS mechanics — player movement and camera control, weapon systems with recoil and reload, hitscan and projectile firing, enemy AI with patrol, chase and attack states, health and damage, HUD and UI, level design, lighting and audio. Written in C# with a component-based architecture; built the full gameplay loop from main menu through win/lose states.

Stack: Unity, C#, Unity NavMesh, Universal Render Pipeline, custom AI state machines.

Agentic Automation — n8n Workflows

Client / Internal · 2025

Multi-step AI agent workflows in n8n Cloud that automate repetitive back-office tasks — pulling data from third-party APIs, running LLM reasoning with guardrails, and writing results back into internal systems. Replaced hours of manual work per week.

Education

B.S. Software Engineering — Mirpur University of Science & Technology (MUST), Mirpur AJK

2022 – 2026

Relevant coursework: Data Structures & Algorithms, Software Architecture, Databases, Artificial Intelligence, Web Technologies, Software Engineering.

Additional

- **Certifications:** [to be added — name, issuer, year]
- **Languages:** English (Professional), Urdu (Native)
- **Availability:** Remote — open to full-time and contract engagements